

Does the legacy of the American frontier make people today less supportive of environmental regulation and collective action to address climate change?

Research Module Macroeconomics
Economic Growth and Gender Dynamics in Historical Perspective

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January 2026

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1 Introduction

By Bastian Rissing

The experience of the American westward-expansion, the so called "American frontier" still shapes today's beliefs about politics and society. What started as a controversial theory about the origins about American "rugged individualism" (Turner, 1893) is still subject of many discussions.¹ But Bazzi et al. (2020) provide a convincing data analysis that proves Turner's thesis. The authors create a new measurement to express the "Total Frontier Experience" (TFE) in years and regress this on different cultural indicators for "rugged individualism" such as the frequency of uncommon names, the preference for the republican party and lower tax rates. They find that more years on the frontier (a higher TFE) actually is a robust indicator for more individualistic political beliefs today.

What we want to extend to this analysis is asking the question whether this frontier experience also shapes beliefs about climate policy today. We differentiate between climate policy opinions on two different types of policy strategies, namely top-down environmental regulation that is passed by the president or the congress and bottom-up collective action that is initiated by regular citizens and local authorities. There are many studies about how political ideology and demography affects opinions on climate change and associated policies. For example we know that conservative white males are more likely to be climate change deniers (McCright and Dunlap, 2011). Regarding the political parties of the US, it is also known that there is a polarization between Democrats and Republicans when it comes to the perception of climate change (Guber, 2013; McCright et al., 2014). Hornsey et al. (2016) conduct a meta-analysis that shows how political ideology is among the strongest predictors in regard to beliefs in climate change, overshadowing education and experience of extreme weather events. When it comes to the motivations behind conservative climate skepticism, Campbell and Kay (2014) show that conservatives do not generally tend to be more climate skeptic because they distrust climate scientists, but because their proposed solutions do not align with their values. Similarly, Boven et al. (2018) argue that Republicans do not disagree about the existence of climate change but that their opposition towards climate change policies simply reflects identity-based and ideological differences with Democrats. Building on this rich literature linking ideology and climate attitudes, we add a historical and cultural dimension by combining these findings with Turner's frontier experience thesis of "rugged individualism". Further studying this possible connection might help in understanding the opposition towards climate policies in some US-regions and the underlying mechanisms and historical roots of climate change scepticism. The results of this study could therefore improve US climate policy design.

¹Commenting on the ethnocentric, colonial and sexist implications of Turner's thesis is out of the scope of this paper but should not stay unmentioned.

We answer this question by regressing the TFE measurement by Bazzi et al. (2020) on various data sets by the Yale Climate Opinion Map on the US-county-level. The Yale Climate Opinion Map consists of over 80 questions regarding opinions about climate change. We distinguish between the two policy approaches (top-down environmental regulation / bottom-up collective action) by using two different questions per approach as the dependent variable, which leads to four questions in total. We also include state fixed effects and geographic control variables to focus on the impact of the TFE.

We also use a mediation analysis and try to separate the influence of the TFE and the influence of republican political ideology on beliefs about climate change. As climate change is a relatively new political topic, it is important to analyze whether counties with frontier experience may be sceptical about climate policy because of their inherent "rugged individualism" that emerged as a result from the frontier experience or because of their preference for the republican party.

2 Theory/Background

By Bastian Rissing

The westward expansion of the United States in the 19th century led to the colonization of land that was previously not governed by a centralized authority and lacked meaningful infrastructure. These regions were marked by low population density and remoteness from more populated areas. According to Turner (1893), these two features shaped the frontier societies which in turn cultivated a strong aversion against any form of intervention by the government and led to the establishment of a strong form of individualism that still influences the regions today. This form of "rugged individualism", a term coined by republican president Herbert Hoover, is an ideology that many people still believe to be an integral part of the "American dream" and is still used by academics and the public to explain differences in political preferences between Americans and their European origins.

2.1 Bazzi et al. (2020)

Bazzi et al. (2020) try to provide empirical evidence for this theory by using econometric methods. To this end, they create a new measurement for expressing the exposure to frontier culture quantitatively on the county-level, the Total Frontier Exposure (TFE). This new key figure shows how many years each county was a part of the American frontier between the years 1790 and 1890 and is calculated by using both population density and closeness to the frontier line (also defined by population density) as indicators². The authors validate these indicators by showing that regions with low population densities show specific demographic

²See Section "Data" for a more precise definition of the TFE.

and cultural traits: their inhabitants are disproportionately young, male, foreign-born, illiterate and more individualistic, which is approximated by having infrequent names. This way, they show that frontier societies indeed had specific cultural traits and were inhabited by more individualistic people in the past.

The study goes on to show that the frontier experience still shapes the affected counties to this day, more than 100 years after the end of classic frontier societies. Higher TFE still correlates with more infrequent names today, which is again used as a proxy for individualism. The authors also show that higher TFE correlates with fiscal conservatism and opposition to redistribution. In practice, this leads to higher voting shares for the republican party and lower property tax rates. This can be seen as evidence for a persistent culture of "rugged individualism" as a result of frontier experience. The authors use different econometric approaches to prove the robustness of their findings, such as additional control variables and an instrumental variable.

At the end, Bazzi et al. (2020) describe the roots of frontier culture. There are two central mechanisms that explain the individualism on the frontier line. The first is selective migration: people that moved to the frontier were already more individualistic than the rest and people that moved away from the frontier into more settled counties were less individualistic than the rest. The second is cultural change: inhabitants that spend more time at the frontier became even more individualistic than before. All in all, the more individualistic inhabitants of the frontier experienced greater economic returns as they were more likely to thrive in the conditions that shaped frontier societies, namely self-reliance and independence.

2.2 Hypotheses

One could argue that the frontier experience can therefore also influence these regions attitudes towards climate change. Bazzi et al. (2020) already show in their research that a higher TFE leads to stronger opposition towards the regulation of CO₂ emissions. But we want to study how exactly this persistence of "rugged individualism" shapes the opposition towards climate policy. We therefore distinguish between two forms of policies that can both be used to combat climate change:

Environmental regulation: We define environmental regulation as laws and taxes that are enforced by a centralized government. In the US this would include the president, congress, the senate and governors. One could also call this approach "top-down" as policies are enforced by a higher political authority.

Collective action: In contrast to that, collective action is a more individualistic approach towards climate change. Instead of centralized governmental choices, actions are coordinated

and taken by citizens and local officials. In contrast to top-down policies this approach can be described as "bottom-up".

If our hypothesis is right and the frontier experience is associated with climate opinions today, we should see that people living in counties with higher TFE-levels oppose both kinds of policy approaches as they are generally sceptical about collective problem-solving, whether as a top-down or a bottom-up approach, especially when the problem concerns a public good such as climate protection. Nevertheless, we should expect a stronger aversion towards top-down environmental regulation than towards bottom-up collective action as a greater TFE leads to greater opposition towards big government than to local solutions. The latter, even though still disliked in general, may still be more popular / less disliked because coordination between regular citizens and local authorities is a more tolerable form of political action for individualistic people. So all in all, we formulate two hypotheses:

1. Higher FTE leads to higher opposition towards climate policies in general.
2. Higher FTE leads to greater opposition towards environmental regulation than towards collective action.

3 Data

By Hannah Arndt and Magaret Vincent

Our empirical analysis combines historical measures of frontier exposure with contemporary county-level data on climate change attitudes in the United States. The unit of observation is the U.S. county, which allows us to exploit fine-grained geographic variation in both historical settlement patterns and modern political beliefs. All datasets are harmonized to consistent county boundaries to ensure comparability across sources and over time. The resulting dataset links counties' historical exposure to frontier conditions during the period of westward expansion to present-day public opinion on climate change and climate policy. After merging all data sources and accounting for missing observations, the final sample consists of 2,036 U.S. counties for which complete information on the dependent variables, the main explanatory variable, and all control variables is available.

3.1 Dependent Variable: YCOM

All outcome variables are taken from the Yale Climate Opinion Maps 2024 (YCOM) dataset. This dataset provides cross-sectional modeled public climate opinion estimates at the local-level across the United States. In the corresponding paper from Howe et al. (2015), the

developers of the YCOM dataset explain how they conducted these estimates. To get these estimates, the dataset employs multilevel regression and poststratification (MRP) (Howe et al., 2015). With this approach Howe et al. (2015) are able to predict climate change beliefs, risk perceptions and policy preferences at the state, congressional district, metropolitan and county levels. According to Howe et al. (2015) this critical social data rarely exist at the sub-national levels and conducting a set of surveys in all of the counties across the United States would be expensive and pooling existing survey data from diverse sources can be problematic. (This is also why we have to use this kind of estimation data?). To address this problem of sparse public opinion data, Howe et al. (2015) use national-scale data to ‘downscale’ estimates of public responses to climate change to sub-national scales. The authors state that given the lack of spatially consistent data on public climate opinions, MRP modelling, when properly optimized and validated, can provide a valuable method for generating estimates or projections at multiple, sub-national geographic scales (Howe et al., 2015). But the margin of error increases as you move to smaller geographic scales. MRP methods can reliably project opinion in areas with sparse data coverage by partially pooling information from survey responses outside of that local geographic unit. But one has to keep in mind that... (possible limitations because of using this kind of data).

Howe et al. (2015) explain that multilevel regression and poststratification (MRP), involves compiling data from multiple national surveys and incorporates demographic, geographic and time variables to partially pool information about respondents across sub-groups. For this, Howe et al. (2015) use Data from 12 nationally representative climate change opinion surveys conducted between 2008 and 2013 for the Yale Project on Climate Change Communication and George Mason Center for Climate Change Communication and merge them into a single combined data set ($n = 12,061$). The MRP process involves two steps. Howe et al. (2015) state that in an initial multilevel regression step, individual survey responses (for example, beliefs, attitudes, policy support, and so on) are modelled as a function of both individual-level demographics (gender, race, education, and a three-way interaction term among these variables) and geography-level covariates (percent of people who drive alone to work, percent of same-sex households, percent of CO2 emissions per capita, and percent of people who voted Democrat in the most recent election). citeHowe2015 include the presidential Democratic vote share as a geographic covariate to marginally improve the model’s descriptive accuracy. They mention that it is possible to drop this term while finding substantively similar results (Howe et al., 2015). But they do not state how much the results change.

During the poststratification stage, the second stage of MRP, Howe et al. (2015) generate a weighted sum of the beliefs of demographic–geographic types for each geographic subunit. That means, Howe et al. (2015) use the multilevel regression model results to project the average opinion of each demographic–geographic-year individual type. As an example, Howe et al. (2015) explain that the model can project the average belief of a Hispanic/Latina

woman with a bachelors degree or higher living in Orange County, California in 2013. The model allows for 32 unique sex–race–education categories, which then interact with 3,143 counties, generating 100,576 unique population types for county-level models (Howe et al., 2015). The census-derived population counts tables provide the count of each population type in each subunit (Howe et al., 2015). Final MRP estimates weight the model-projected belief of each population type by the true population count of that type in a given geographic subunit (Howe et al., 2015).

Our analysis uses two public climate change opinion variables capturing the top down environmental regulation approach and two other ones capturing the bottom up collective action approach, both drawn from YCOM. Each of these outcome variables is defined as the estimated share of adults in area i who agree with that item and is therefore expressed on a 0-100 percentage scale.

The following two public climate change opinion variables are chosen to represent environmental regulations:

1. Estimated percentage who think citizens should do less to address global warming
The Interpretation of this would be the following. The “Estimated percentage who think citizens should do less to address global warming” represents the poststratified estimate of support for this view, calculated by projecting the expected belief for each demographic–geographic–year population type from the multilevel regression and weighting and averaging these projections by the true population counts within the geographic unit.
2. Estimated percentage who think their local officials should do less to address global warming

The following two public climate change opinion variables are chosen to represent collective action:

1. Estimated percentage who do not say global warming should be a high priority for the next president and Congress
2. Estimated percentage who oppose requiring fossil fuel companies to pay a carbon tax and use the money to reduce other taxes (such as income tax) by an equal amount

In different regressions, each of these YCOM variables is used respectively as a dependent variable, allowing the analysis to measure how Total Frontier Exposure relates to different dimensions of climate attitudes.

3.2 Main Regressor: TFE

Our main explanatory variable is Total Frontier Experience (TFE), developed by Bazzi, Fiszbein, and Gebresilasse (2020). The TFE measures the cumulative number of years a county was exposed to frontier conditions during the period of U.S. westward expansion, between 1790 and 1890.

Following the definition proposed by Turner and formalized in the U.S. Census Bureau’s 1890 report, we define the frontier line as the boundary at which population density falls below two people per square mile (Bazzi et al., 2020). The frontier is thus composed of counties characterized by low population density and located in close geographic proximity to this boundary. Based on this time-varying classification, Total Frontier Experience (TFE) is constructed as the cumulative number of years a county spent under frontier conditions between 1790 and 1890. Consequently, counties that were settled later or remained sparsely populated for longer periods exhibit higher TFE values (Bazzi et al., 2020).

Turner conceptualized frontier as a "form of society" rather than an area, by identifying characteristics such as isolation and abundance of resources. The TFE is designed to capture prolonged exposure to frontier conditions characterized by low population density, weak state capacity, limited infrastructure, and high reliance on individual self-sufficiency. Bazzi et al. (2020) show that this historical exposure is strongly associated with persistent cultural traits, including political individualism, opposition to government intervention, and skepticism toward collective solutions.

To measure the duration of frontier exposure for each county, we follow the methodology developed by Bazzi, Fiszbein, and Gebresilasse (2020). Specifically, we calculate the number of years each county spent in the frontier belt between 1790 and 1890 using a binary indicator for each year that denotes whether the county met the frontier criteria defined by proximity and low density. The Total Frontier Experience (TFE) for a county is then defined as the sum of these annual indicators. This approach allows us to quantify historical frontier exposure in a manner consistent with Bazzi et al. (2020), capturing variation in the intensity and duration of frontier conditions across counties.

3.3 Controls

Following Bazzi et al. (2020), our regressions include Geographic/Agroclimatic Controls to absorb fixed spatial confounders. Meant are the following variables: county area; county centroid latitude and longitude; distance to oceans, lakes, rivers and portage sites from county centroid; mean county temperature and rainfall; elevation; and average potential agricultural yield and ruggedness. Our control strategy also follows Bazzi et al. (2020) in such a way that, we also incorporate historical controls. These include the 1890 sex ratio, 1890 birthplace diversity, 1890 birthplace diversity, 1890 manufacturing employment share, 1890

immigrant share, years connected to railroad by 1890, 1860 slave population share, 1890 scottish irish immigrant share and distance to nearest mineral discovery site and Indian battle. We include the same set of historical controls to adress the concern that TFE may be correlated with persistent, unobserved features of local development, that could independently shape contemporary climate attitudes.

In addition, climate change perceptions possibly exhibit geographic patterns due to differences in personal experience with extreme weather events and climate variability, since local weather is known to influence climate change perceptions. Thats why we also include hurricanes, tornados. . . .?

3.4 Voting data

4 Empirical Strategy

By Saskia Schatzmayr

This section explains how we estimate the relationship between historical frontier exposure and today’s climate-policy attitudes. We use four county-level outcomes from the Yale Climate Opinion Maps (YCOM), focusing on opposing questions (e.g. estimated population share who oppose requiring fossil fuel companies to pay a carbon tax) or negatively worded questions (e.g. estimated population share who think citizens should do less to address global warming) to capture climate-policy opposition. This choice aligns with our theoretical focus on opposition to collective problem-solving and skepticism toward climate policy and allows us to investigate both hypotheses, the collective action (*bottom-up*), and the environmental regulation (*top-down*).

To do so, we proceed in three steps. First, we replicate the central relationship emphasized in Bazzi et al. (2020) by estimating the effect of Total Frontier Experience (TFE) on present-day Republican vote share. Second, we estimate whether TFE is associated with climate-policy attitudes today. Third, motivated by the strong role of political ideology in climate attitudes, we explore whether the relationship between TFE and climate attitude (a) operates through contemporary political ideology using diagnostic mediation analysis decomposition and (b) is heterogeneous across the partisan environment using a moderation analysis with an interaction term.

Let i index counties and $s(i)$ denote the state of county i . Throughout, all regressions are estimated at the county level and include the full set of geographic controls G_i , historical controls H_i , and state fixed effects $\delta_{s(i)}$. Standard errors are clustered at the 60km grid-cell

level ³.

4.1 Step 1: Replicating the TFE and partisanship relationship

We start by replicating one of the results highlighted in Bazzi et al. (2020). Let TFE_i denote county i 's total frontier experience in years, and let R_i denote county i 's Republican vote share in the 2020 presidential election (scaled 0–1).

We estimate:

$$R_i = \alpha_0 + \alpha_1 TFE_i + G_i' \alpha_2 + H_i' \alpha_3 + \delta_{s(i)} + u_i. \quad (1)$$

Equation (1) establishes whether higher frontier exposure is associated with higher contemporary Republican vote share, consistent with the persistence effect documented by Bazzi et al. (2020).

4.2 Step 2: Frontier exposure and climate-policy opposition

Next, we estimate whether frontier exposure is associated with climate-policy opposition today. Let y_i denote a county-level YCOM outcome (in percentage points, 0–100) for one of the four opposing questions. Our baseline specification is:

$$y_i = \beta_0 + \beta_1 TFE_i + G_i' \beta_2 + H_i' \beta_3 + \delta_{s(i)} + \varepsilon_i. \quad (2)$$

We estimate equation (2) for each of the four outcomes, and compare results across the two policy domains (collective action versus environmental regulation) to assess whether “rugged individualism” is more strongly related to opposition toward top-down regulation than to bottom-up action.

4.3 Step 3a: Diagnostic Mediation-style decomposition using Republican vote share

A central question is whether the association between frontier exposure and climate attitude reflects frontier culture per se or whether it is statistically intertwined with contemporary partisan preferences. To explore this, we include Republican vote share in our baseline regression:

$$y_i = \gamma_0 + \gamma_1 TFE_i + \gamma_2 R_i + G_i' \gamma_3 + H_i' \gamma_4 + \delta_{s(i)} + \eta_i. \quad (3)$$

Comparing β_1 from (2) with γ_1 from (3) is a descriptive decomposition: it shows how much of the baseline TFE coefficient is absorbed once we condition on county partisanship.

³This is important because counties in geographic proximity may share unobserved regional determinants of climate attitudes (e.g. political culture, economic structure, or regional media). Such correlation could otherwise understate standard errors and overstate statistical significance.

Identification limits for the mediation We emphasize that equation (3) should *not* be interpreted as identifying a causal mediation channel through R_i . The YCOM county-level opinion estimates are generated using multilevel modeling and poststratification, and the estimation procedure includes electoral vote-share covariates to improve predictive accuracy. As a result, including R_i on the right-hand side can introduce a mechanical relationship between R_i and the dependent variable y_i . For this reason, we treat the “mediation analysis” as a diagnostic robustness and descriptive decomposition rather than a causal mediation analysis. This means, we do not interpret γ_2 as the causal effect of Republican vote share on climate attitudes, nor do we interpret the change from β_1 to γ_1 as identifying a causal mediation channel.

4.4 Step 3b: Moderation analysis using an partisanship interaction term

Following the decomposition above, we also test whether the relationship between frontier exposure and climate attitudes is heterogeneous across the partisan environment. Conceptually, this is a moderation (heterogeneity) test and we ask whether frontier exposure is more strongly associated with climate-policy opposition counties with a higher Republican vote share. For interpretability of main effects, we center both variables, TFE_i and R_i around their mean ⁴:

$$TFE_i^c = TFE_i - \overline{TFE}, \quad R_i^c = R_i - \bar{R}.$$

and estimate:

$$y_i = \mu_0 + \mu_1 TFE_i^c + \mu_2 R_i^c + \mu_3 (TFE_i^c \times R_i^c) + G_i' \mu_4 + H_i' \mu_5 + \delta_{s(i)} + \xi_i. \quad (4)$$

The interaction coefficient μ_3 captures whether the marginal effect of TFE with y varies with county Republican vote share. The marginal effect of TFE implied by (4) is:

$$\frac{\partial y_i}{\partial TFE_i} = \mu_1 + \mu_3 R_i^c.$$

As in the mediation-style specification, this is primarily a descriptive moderation analysis. The presence of vote-share covariates in the construction of YCOM outcomes estimates requires caution when interpreting relationships involving R_i .

Robustness: Controlling for contemporary natural-disaster risk or different survey questions.

⁴Centering does not change the model but rather shifts the reference point for main effects so that μ_1 reflects the relationship of TFE with y at average Republican vote share (and analogously for μ_2).

4.5 Identification of causal effects

The inclusion of historical and geographic controls as well as state fixed effects in our models reduces the effect of observable and state-level unobservable determinants. Additionally, clustering standard errors reduces the effect of unobserved regional determinants of counties with close proximity. However, we do not claim full causal identification of the effect of frontier exposure on climate attitudes. Remaining threats include county-level unobservable variables correlated with both TFE_i and climate attitudes, and the fact that YCOM outcomes are generated estimates rather than directly observed population shares. Moreover, specifications including R_i face an additional limitation because vote share is used in the construction of YCOM county-level estimates. Coefficients on R_i may be inflated and standard errors may be optimistic. We therefore interpret our findings as robust conditional associations and use the mediation and moderation models as diagnostic tools rather than causal mechanism tests.

5 Results

By Saskia Schatzmayr

5.1 Replication: TFE and Republican vote share

We reproduce the positive relationship between frontier exposure and contemporary Republican vote share highlighted by Bazzi et al. (2020), shown in Table 1. The estimated coefficient on TFE is 0.018 (0.003), which implies that one additional year of frontier experience is associated with about 1.8 percentage points higher Republican vote share, conditional on controls and state fixed effects.

Table 1: Mediator regression: Frontier Experience and Republican Vote Share (2020)

	β	σ	t	p	$ci(95\%)$
Total Frontier Experience (years)	0.018	(0.003)	5.04	0.000	0.011,0.024
Observations	2036				
R-squared	0.513				

5.2 Environmental Regulation

The results of the baseline regression for the environmental regulations are moderate yet significant. In panel A of the Table 2 and Table 3, the coefficient on TFE is positive and statistically significant for both outcomes. For `priorityOppose`, the estimate is 0.486 (0.135);

for `reducetaxOppose`, it is 0.535 (0.131). Since the dependent variables are measured in percentage points, these estimates imply that a 1-year increase in frontier exposure is associated with roughly 0.49 to 0.54 percentage points higher opposition to top-down climate policy, holding the controls and state fixed effects constant.

Table 2: Estimated percentage who do not say global warming should be a high priority for the next president and Congress

	β	σ	t	p
<i>Panel A: Baseline</i>				
TFE (years)	0.486	(0.135)	3.61	0.000
Observations	2036			
R-squared	0.465			
<i>Panel B: Mediation</i>				
TFE (years)	-0.054	(0.065)	-0.82	0.411
Republican vote share (2020)	30.695	(0.537)	57.18	0.000
Observations	2036			
R-squared	0.820			
<i>Panel C: Moderation</i>				
TFE (centered)	-0.051	(0.065)	-0.78	0.437
RepShare (centered)	30.706	(0.529)	58.04	0.000
TFE (centered) $\times$ RepShare (centered)	0.940	(0.318)	2.96	0.003
Observations	2036			
R-squared	0.821			

Panel B of each table adds county Republican vote share. In both top-down outcomes, the TFE coefficient becomes small and statistically insignificant from zero (-0.054 for `priorityOppose` and -0.047 for `reducetaxOppose`). At the same time, the coefficient on `rep_share` is large and very precisely estimated (around 30–33 percentage points), and the R^2 increases sharply. We emphasize the key limitation discussed in the empirical strategy: YCOM outcomes are model-based estimates that use vote-share covariates in their construction.

Panel C reports the moderation regressions with centered variables. In both cases, the interaction term between centered TFE and centered Republican vote share is positive and statistically significant. This indicates that the relationship between TFE and climate opposition is stronger in counties with higher Republican vote share (relative to the mean).

5.3 Collective Action

In Panel A of Tables 4 and 5, the TFE coefficient is positive and statistically significant for both outcomes. The point estimate is 0.239 (0.061) for `localofficialsOppose` and 0.249 (0.049) for `citizensOppose`. This implies that a 1-year increase in frontier exposure

Table 3: Estimated percentage who oppose requiring fossil fuel companies to pay a carbon tax and use the money to reduce other taxes (such as income tax) by an equal amount

	β	σ	t	p
<i>Panel A: Baseline</i>				
TFE (years)	0.535	(0.131)	4.08	0.000
Observations	2036			
R-squared	0.506			
<i>Panel B: Mediation</i>				
TFE (years)	-0.047	(0.045)	-1.05	0.297
Republican vote share (2020)	33.113	(0.480)	68.96	0.000
Observations	2036			
R-squared	0.910			
<i>Panel C: Moderation</i>				
TFE (centered)	-0.045	(0.045)	-1.01	0.312
RepShare (centered)	33.121	(0.480)	69.06	0.000
TFE (centered) $\times$ RepShare (centered)	0.698	(0.265)	2.64	0.009
Observations	2036			
R-squared	0.910			

is associated with roughly 0.24 to 0.25 percentage points higher opposition to bottom-up climate action, conditional on the controls and state fixed effects.

As with the top-down outcomes, adding `rep_share` in Panel B reduces the estimated TFE coefficient. For `localofficialsOppose`, the TFE coefficient becomes essentially zero (0.003). For `citizensOppose`, it falls to 0.049 and is only marginally significant (at the 10% level). Republican vote share remains very strongly associated with both outcomes, and R^2 rises sharply. These patterns again point to a tight empirical connection between partisanship and climate-policy opposition, but they should be treated as descriptive because vote-share covariates enter the YCOM estimation procedure.

Tables 4 and 5 show positive and statistically significant interaction terms for both bottom-up outcomes. This implies that the association between frontier exposure and opposition to bottom-up action is stronger in more Republican counties (relative to the mean), consistent with the results from the top-down approach.

Table 4: Estimated percentage who think their local officials should do less to address global warming

	β	σ	t	p
<i>Panel A: Baseline</i>				
TFE (years)	0.239	(0.061)	3.95	0.000
Observations	2036			
R-squared	0.490			
<i>Panel B: Mediation</i>				
TFE (years)	0.003	(0.031)	0.08	0.934
Republican vote share (2020)	13.464	(0.257)	52.37	0.000
Observations	2036			
R-squared	0.825			
<i>Panel C: Moderation</i>				
TFE (centered)	0.004	(0.031)	0.13	0.899
RepShare (centered)	13.469	(0.258)	52.18	0.000
TFE (centered) $\times$ RepShare (centered)	0.436	(0.126)	3.47	0.001
Observations	2036			
R-squared	0.826			

Table 5: Estimated percentage who think citizens should do less to address global warming

	β	σ	t	p
<i>Panel A: Baseline</i>				
TFE (years)	0.249	(0.049)	5.05	0.000
Observations	2036			
R-squared	0.515			
<i>Panel B: Mediation</i>				
TFE (years)	0.049	(0.026)	1.89	0.060
Republican vote share (2020)	11.350	(0.220)	51.69	0.000
Observations	2036			
R-squared	0.818			
<i>Panel C: Moderation</i>				
TFE (centered)	0.050	(0.026)	1.93	0.055
RepShare (centered)	11.354	(0.220)	51.65	0.000
TFE (centered) $\times$ RepShare (centered)	0.352	(0.124)	2.85	0.005
Observations	2036			
R-squared	0.819			

5.4 Meaning for Hypotheses

6 Implications

7 Conclusion

- questions maybe too inaccurate for our purposes - problem: having estimates (YCOM) as dependent variable

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Appendix

Table 6: Estimated percentage who oppose requiring fossil fuel companies to pay a carbon tax and use the money to reduce other taxes (such as income tax) by an equal amount (with NRI)

	β	σ	t	p	ci95%
<i>Panel A: Outcome</i>					
Total Frontier Experience (years)	0.402	(0.132)	3.03	0.003	0.141, 0.662
Observations	2036				
R-squared	0.537				
<i>Panel B: Outcome + RepShare</i>					
Total Frontier Experience (years)	-0.020	(0.044)	-0.45	0.655	-0.107, 0.067
Republican vote share, county total (2020)	34.362	(0.524)	65.60	0.000	33.332, 35.392
Observations	2036				
R-squared	0.913				

Dependent variable: reducetaxOppose.

All models include geographic and historical controls (including NRI) and state fixed effects.

Standard errors clustered at 60km grid cell (km_grid_cel_code).

Note: RepShare is included in the YCOM estimator; interpret Panel B as diagnostic, not causal mediation.

Table 7: Estimated percentage who do not say global warming should be a high priority for the next president and Congress (with NRI)

	β	σ	t	p	ci95%
<i>Panel A: Outcome</i>					
Total Frontier Experience (years)	0.372	(0.137)	2.71	0.007	0.102, 0.642
Observations	2036				
R-squared	0.489				
<i>Panel B: Outcome + RepShare</i>					
Total Frontier Experience (years)	-0.022	(0.064)	-0.35	0.729	-0.149, 0.104
Republican vote share, county total (2020)	32.106	(0.570)	56.33	0.000	30.986, 33.227
Observations	2036				
R-squared	0.825				

Dependent variable: priorityOppose.

All models include geographic and historical controls (including NRI) and state fixed effects.

Standard errors clustered at 60km grid cell (km_grid_cel_code).

Note: RepShare is included in the YCOM estimator; interpret Panel B as diagnostic, not causal mediation.

Table 8: Estimated percentage who oppose requiring fossil fuel companies to pay a carbon tax and use the money to reduce other taxes (such as income tax) by an equal amount

	β	σ	t	p	ci95%
<i>Panel A: Outcome</i>					
Total Frontier Experience (years)	0.535	(0.131)	4.08	0.000	0.277, 0.792
Observations	2036				
R-squared	0.506				
<i>Panel B: Outcome + RepShare</i>					
Total Frontier Experience (years)	-0.047	(0.045)	-1.05	0.297	-0.137, 0.042
Republican vote share, county total (2020)	33.113	(0.480)	68.96	0.000	32.169, 34.057
Observations	2036				
R-squared	0.910				

Dependent variable: `reducesTaxOppose`.

All models include geographic and historical controls and state fixed effects.

Standard errors clustered at 60km grid cell.

Note: `RepShare` is included in the YCOM estimator; interpret Panel B as diagnostic, not causal mediation.

Table 9: Estimated percentage who do not say global warming should be a high priority for the next president and Congress

	β	σ	t	p	ci95%
<i>Panel A: Outcome</i>					
Total Frontier Experience (years)	0.486	(0.135)	3.61	0.000	0.221, 0.751
Observations	2036				
R-squared	0.465				
<i>Panel B: Outcome + RepShare</i>					
Total Frontier Experience (years)	-0.054	(0.065)	-0.82	0.411	-0.182, 0.075
Republican vote share, county total (2020)	30.695	(0.537)	57.18	0.000	29.639, 31.750
Observations	2036				
R-squared	0.820				

Dependent variable: `priorityOppose`.

All models include geographic and historical controls and state fixed effects.

Standard errors clustered at 60km grid cell.

Note: `RepShare` is included in the YCOM estimator; interpret Panel B as diagnostic, not causal mediation.

Table 10: Estimated percentage who think their local officials should do less to address global warming

	β	σ	t	p	ci95%
<i>Panel A: Outcome</i>					
Total Frontier Experience (years)	0.239	(0.061)	3.95	0.000	0.120, 0.358
Observations	2036				
R-squared	0.490				
<i>Panel B: Outcome + RepShare</i>					
Total Frontier Experience (years)	0.003	(0.031)	0.08	0.934	-0.058, 0.063
Republican vote share, county total (2020)	13.464	(0.257)	52.37	0.000	12.958, 13.969
Observations	2036				
R-squared	0.825				

Dependent variable: localofficialsOppose.

All models include geographic and historical controls and state fixed effects.

Standard errors clustered at 60km grid cell.

Note: RepShare is included in the YCOM estimator; interpret Panel B as diagnostic, not causal mediation.

Table 11: Estimated percentage who think citizens should do less to address global warming

	β	σ	t	p	ci95%
<i>Panel A: Outcome</i>					
Total Frontier Experience (years)	0.249	(0.049)	5.05	0.000	0.152, 0.346
Observations	2036				
R-squared	0.515				
<i>Panel B: Outcome + RepShare</i>					
Total Frontier Experience (years)	0.049	(0.026)	1.89	0.060	-0.002, 0.101
Republican vote share, county total (2020)	11.350	(0.220)	51.69	0.000	10.918, 11.782
Observations	2036				
R-squared	0.818				

Dependent variable: citizensOppose.

All models include geographic and historical controls and state fixed effects.

Standard errors clustered at 60km grid cell.

Note: RepShare is included in the YCOM estimator; interpret Panel B as diagnostic, not causal mediation.

Table 12: Estimated percentage who think their local officials should do less to address global warming (with NRI)

	β	σ	t	p	ci95%
<i>Panel A: Outcome</i>					
Total Frontier Experience (years)	0.206	(0.062)	3.32	0.001	0.084, 0.328
Observations	2036				
R-squared	0.499				
<i>Panel B: Outcome + RepShare</i>					
Total Frontier Experience (years)	0.027	(0.030)	0.91	0.362	-0.031, 0.086
Republican vote share, county total (2020)	14.577	(0.268)	54.35	0.000	14.050, 15.104
Observations	2036				
R-squared	0.840				

Dependent variable: localofficialsOppose.

All models include geographic and historical controls (including NRI) and state fixed effects.

Standard errors clustered at 60km grid cell (km_grid_cel_code).

Note: RepShare is included in the YCOM estimator; interpret Panel B as diagnostic, not causal mediation.

Table 13: Estimated percentage who think citizens should do less to address global warming (with NRI control)

	β	σ	t	p	ci95%
<i>Panel A: Outcome</i>					
Total Frontier Experience (years)	0.211	(0.050)	4.24	0.000	0.113, 0.309
Observations	2036				
R-squared	0.531				
<i>Panel B: Outcome + RepShare</i>					
Total Frontier Experience (years)	0.064	(0.026)	2.43	0.016	0.012, 0.115
Republican vote share, county total (2020)	11.996	(0.248)	48.46	0.000	11.509, 12.483
Observations	2036				
R-squared	0.824				

Dependent variable: citizensOppose.

All models include geographic and historical controls (including NRI) and state fixed effects.

Standard errors clustered at 60km grid cell (km_grid_cel_code).

Note: RepShare is included in the YCOM estimator; interpret Panel B as diagnostic, not causal mediation.